

AUTOMATE YOUR HOME WITH ZigBee

Set up a low-cost wireless sensor network for automatic control of fans, lights and other appliances in your home



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The last decade witnessed development and implementation of various wireless communication standards. GSM, WiFi and Bluetooth are well known amongst them. These standards have penetrated into our daily routine and seem to have fulfilled wireless requirements of all people. However, something like an Internet of things is amiss, especially in mainstream home automation (HA).

HA systems provide interoperability between various electronic, electrical and power devices as well as an inter-

active interface to control their operation. These systems help in optimising and economising the energy consumption. They also make life easier, especially for the elderly and physically disabled.

Most available HA systems are non-interoperable, expensive and often wired systems. Wiring complicates implementation of home automation in buildings that are already built. A relatively new wireless standard called ZigBee has emerged to solve this problem. It allows for an 'open' and standardised wireless network of battery-powered low-cost sensors, actuators and control devices that could effectively communicate with each other. As the name indicates, it involves packet routing strategies in networks with large

nodes and zig-zag transmission of data.

ZigBee applications, markets and forecasts

Though development of the ZigBee standard is still under progress, there are lot many applications for ZigBee. The most promising are:

1. Home control: Security, heating, ventilation and air-conditioning (HVAC), lighting control, access control, irrigation
2. Personal health care: Patient monitoring, fitness monitoring
3. Industrial control: Asset management, process control, energy man-

Advantages of ZigBee

1. Chip-vendor independence
2. Rapid innovation
3. Extremely low cost
4. Open standards enable markets
5. Easy to deploy
6. Excellent performance in environments with low signal-to-noise ratio
7. Appropriate range of operation (30-100 metres)
8. Appropriate and reliable data transfer, bitrate: 250 kbps (at 2.4 GHz)
9. Very low power consumption
10. Secure data transfer
11. ZigBee protocol needs less than 64 kb of ROM and 2-32 kb of RAM
12. ZigBee can be implemented with any type of microcontroller

- agement, environmental
4. Building automation: Automatic meter reading (AMR), security, HVAC, lighting control, access control
5. Consumer electronics: Remote control
6. PCs and peripherals: Mouse, keyboard, joystick
7. Environment: Environment monitoring

Network reference model

Network devices, whether wired or wireless, are commonly described by the open systems interconnection (OSI) reference model. The generic seven-layer model is applied to all network and media types.

Fig. 1 shows the ISO-OSI network reference model for ZigBee purposes.